

Project Title :

Text Steganography System Using Python (GUI) .

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Class :

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Subject :

Information Security

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Introduction :-

This project is based on Text Steganography. In this system, a message is converted into encoded form and then decoded back to the original message. It helps to understand basic data hiding techniques using Python GUI.

Objective :-

The main objective of this project is to:

- Learn message encoding and decoding
- Understand basic data hiding concept
- Create a simple GUI application using Python
- Improve programming skills

Tools and Technologies :-

- Python Programming Language
- Tkinter (for GUI)
- Base64 Module

Working of Project :-

In this project:

- The user enters a message in the input box.
- When the Encode button is clicked, the message is converted into encoded form using Base64.
- The encoded message is shown in the output box.
- When the Decode button is clicked, the original message is recovered.

Modules :-

- Input Module (for entering message)
- Encoding Module (convert message into encoded form)
- Decoding Module (convert back to original message)
- GUI Module (for user interface)

Algorithm :-

1 . Encoding Process :

- Start
- Enter message
- Convert message into Base64 format
- Display encoded message
- Stop

2 . Decoding Process:

- Start
- Enter encoded message
- Convert it back to original message
- Display result
- Stop

Advantages :-

- Easy to use
- Simple GUI interface
- Helps to understand encoding concept
- Fast processing

Limitations :-

- This is not real steganography
- It is only encoding and decoding
- Security level is low

Future Improvements :-

- Add image steganography
- Add encryption (AES)
- Add login system
- Improve security level

Conclusion :-

This project helps to understand basic data hiding techniques using encoding. It also improves knowledge of Python GUI programming and basic security concepts.